

T3 matched EFT Lagrangian terms: EFT_1_after_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

Field combinations 1 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1 D$	S_2
								$+C_{S_1^\dagger S_1} S_{1,j_1} S_{1,j_1}^\dagger$		$+C_{S_2^\dagger S_2} S_{2,j_1} S_{2,j_1}^\dagger$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{1}{2} C_{S_1^\dagger S_1} \overline{h y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger$	$+D_{\mu_1} (S_{1,j_1})^\dagger D_{\mu_1} (S_{1,j_1})$	$-\frac{1}{2} C_{S_2^\dagger S_2} \overline{h y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger$
								$\frac{C_{S_1^\dagger S_1} \overline{h y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger}{\epsilon}$		$\frac{C_{S_2^\dagger S_2} \overline{h y_{2,i_1,r_1}} S_{2,j_1} y_{2,i_1,r_1} S_{2,j_1}^\dagger}{\epsilon}$
								$-\frac{\overline{h y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{\epsilon}$		$-\frac{\overline{h y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$
cont.								$+ \overline{h y_{1,i_1,r_1} y_{1,i_2,r_1}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} S_{1,j_1}^\dagger S_{1,j_2}^\dagger$		$+ \overline{h y_{2,i_1,r_1} y_{2,i_2,r_1}} S_{2,j_1} S_{2,j_2} y_{2,i_1,r_2} y_{2,i_2,r_2} S_{2,j_1}^\dagger S_{2,j_2}^\dagger$
								$+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}$		$+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}$
								$-\frac{\overline{h y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}}{\epsilon}$		$-\frac{\overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2} \overline{h y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^{(1)}$		$-\frac{1}{2} \overline{h y_{2,i_1,r_1}} S_{2,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^{(1)}$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 D$	$H \ell$	$S_1 H$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1} (S_{2,j_1})^\dagger D_{\mu_1} (S_{2,j_1})$	$+\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}e M_F}$ $+\frac{2h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}\overline{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2h\overline{\lambda_{T3}}\overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}e M_F}$	$+\frac{2h\overline{y_{2,i_1,r_1}}\overline{H_{a_1}} H_{a_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} \lambda_{T3} \overline{\mathcal{I}_{y2,a_1,j_1}}^* \overline{\mathcal{I}_{y2,a_2,j_2}}^*}{\sqrt{3}e}$ $+\frac{h\overline{y_{2,i_1,r_1}}\overline{H_{a_1}} H_{a_2} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} \lambda_{T3} \overline{\mathcal{I}_{y2,a_1,j_1}}^* \overline{\mathcal{I}_{y2,a_2,j_2}}^*}{\sqrt{3}}$ $-\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \overline{\mathcal{I}_{y1,a_1,j_2}}^* \overline{\mathcal{I}_{y1,a_2,j_1}}^* H_{a_1}^\dagger S_{1,j_1}^\dagger}{\epsilon}$ $-\frac{2h\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \overline{\mathcal{I}_{y1,a_1,j_2}}^* \overline{\mathcal{I}_{y1,a_2,j_1}}^* H_{a_1}^\dagger S_{1,j_1}^\dagger}{\sqrt{3}e}$ $+\frac{h\overline{y_{1,i_1,r_1}}\overline{\lambda_{T3}} y_{2,i_1,r_1} \overline{\mathcal{I}_{y2,a_1,j_2}} \overline{\mathcal{I}_{y2,a_2,j_1}} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger S_{1,j_2}^\dagger}{\sqrt{3}}$ $+H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^1 \lambda_{H1}^{(1)}$
cont.									$+\frac{2h\overline{\lambda_{T3}}\overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$-\frac{h\overline{y_{1,i_1,r_1}} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^1 \lambda_{H1}^{(1)}}{\epsilon}$
cont.										$-\frac{1}{2} \frac{h\overline{y_{1,i_1,r_1}} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^1 \lambda_{H1}^{(1)}}{h\overline{y_{1,i_1,r_1}} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^2 \lambda_{H1}^{(2)}}$
cont.										$-\frac{1}{2} \frac{h\overline{y_{1,i_1,r_1}} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^2 \lambda_{H1}^{(2)}}{h\overline{y_{1,i_1,r_1}} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger \overline{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^2 \lambda_{H1}^{(2)}}$

Field combinations 3 of 8

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2$	$S_1 F$	$S_1 \ell$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+C \text{NewScalar1 NewScalar2} h S_{1,j_1} S_{2,j_2} \text{Bar}(\text{cg1})_{j_1,j_2}$ $+ \frac{1}{2} C S_2^\dagger S_2 \overline{h y_{2,i_1,r_1}} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg1})_{j_1,j_2}$ $+ \frac{C S_2^\dagger S_2 \overline{h y_{2,i_1,r_1}} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg1})_{j_1,j_2}}{\epsilon}$	$-C_{S_1^\dagger S_1} h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}^\dagger$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_2} S_{1,j_1}^\dagger S_{1,j_2}^\dagger$ $-h \log \left(\frac{\tilde{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}^\dagger S_{1,j_2}^\dagger} I_{S_{1,j_1,j_3,j_4}} \lambda_{S_1}^{(1)}$	$\frac{3h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_1} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}} g_2^2}}{8\epsilon M_F}$ $+ \frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_1} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}} g_2^2}}{8M_F}$ $+ \frac{3h \overline{y_{1,i_1,r_1} y_{1,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}} S_{1,j_1}^\dagger S_{1,j_2}^\dagger} g_2^2}}{8\epsilon M_F}$
cont.								$- \frac{h \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_1} S_{1,j_2} S_{2,j_3} y_{1,i_1,r_2} y_{1,i_2,r_2} \text{Bar}(\text{cg1})_{j_2,j_3} \text{Bar}(\text{cg1})_{j_3,j_4}}{\epsilon}$ $+ h \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_1} S_{1,j_2} S_{2,j_3} S_{2,j_4} y_{1,i_1,r_2} y_{1,i_2,r_2} \text{Bar}(\text{cg1})_{j_1,j_2} \text{Bar}(\text{cg1})_{j_3,j_4}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_1} S_{1,j_2} S_{2,j_3} y_{1,i_1,r_2} y_{1,i_2,r_2} \text{Bar}(\text{cg1})_{j_2,j_3} S_{2,j_4}^\dagger}{\epsilon}$		$+ \frac{h \overline{y_{1,i_1,r_1} y_{1,i_2,r_1}} L_{a_2,i_2}^T C P_L L_{a_1,i_1} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}} S_{1,j_1}^\dagger S_{1,j_2}^\dagger} g_2^2}}{8M_F}$ $+ \frac{3h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_1} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}}}}{8\epsilon M_F}$ $+ \frac{h \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_1} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}}}}{8M_F}$ $+ \frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_1} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}}}}{2M_F}$
cont.								$- \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_1} S_{1,j_2} S_{2,j_3} y_{1,i_1,r_2} y_{1,i_2,r_2} \text{Bar}(\text{cg1})_{j_2,j_3} S_{1,j_1}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{2,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,i_1,r_2} y_{2,i_2,r_2} \text{Bar}(\text{cg1})_{j_2,j_3} S_{2,j_4}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{2,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} S_{2,j_3} y_{1,i_1,r_2} y_{2,i_2,r_2} \text{Bar}(\text{cg1})_{j_2,j_3} S_{2,j_4}^\dagger}{\epsilon}$ $+ h C \text{NewScalar1 NewScalar2} \text{cg1}_{j_1,j_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$		$- \frac{h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_2} y_{1,i_3,r_2} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}}}}{4\epsilon M_F}$ $- \frac{h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_2} y_{1,i_3,r_2} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}}}}{8M_F}$ $- \frac{h \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_3,r_2} y_{1,i_3,r_2} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}}}}{4M_F}$
cont.								$+ \frac{1}{2} C S_2^\dagger S_2 \overline{h y_{1,i_1,r_1}} \text{cg1}_{j_1,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger$ $+ \frac{C S_2^\dagger S_2 \overline{h y_{1,i_1,r_1}} \text{cg1}_{j_1,j_2} y_{2,i_1,r_1} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$		$- \frac{h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_2} y_{1,i_3,r_2} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}}}}{4\epsilon M_F}$ $- \frac{3h \overline{y_{1,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_1} y_{1,i_2,r_2} y_{1,i_3,r_2} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}}}}{8M_F}$ $- \frac{h \overline{y_{2,i_3,r_1}} L_{a_1,i_1} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{2,i_3,r_2} y_{1,i_3,r_2} I_{y_{1,a_1,j_1} T_{y_{1,a_2,j_2}}}}{4\epsilon M_F}$
cont.								$+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $- \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $- \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$ $+ \frac{2h \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_2} y_{1,i_1,r_2} S_{1,j_1}^\dagger S_{2,j_2}^\dagger}{\epsilon}$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2$	$S_1 F$	$S_1 \ell$
								$-\frac{1}{2}\sqrt{3h\overline{y_{1,i_1,r_1}}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$		
cont.								$+\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}}{\epsilon}$		
								$+\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$		
								$-\frac{\sqrt{3h\overline{y_{2,i_1,r_1}}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}}{\epsilon}$		
cont.								$-\frac{1}{2}\sqrt{3h\overline{y_{2,i_1,r_1}}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\lambda_{12}^{(2)}$		
								$+\frac{2h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger\lambda_{12}^{(2)}}{\epsilon}$		
								$+\frac{h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{1,j_2}^\dagger S_{2,j_3}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}}$		
cont.								$+\frac{S_{1,j_3}S_{2,j_4}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$		
								$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$		
								$-\frac{1}{2}h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$		
cont.								$-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$		
								$-\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$		
								$+\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}1)_{j_2,j_3}S_{2,j_1}^\dagger\lambda_{S_2}^{(1)}}{\epsilon}$		
cont.								$+\frac{1}{2}h\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\text{Bar}(\text{cg}1)_{j_2,j_3}S_{2,j_1}^\dagger\lambda_{S_2}^{(1)}$		
								$+\frac{h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger\lambda_{S_2}^{(1)}}{\epsilon}$		
cont.								$+\frac{1}{2}h\overline{y_{1,i_1,r_1}}\text{cg}1_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}^\dagger S_{2,j_2}^\dagger S_{2,j_3}^\dagger\lambda_{S_2}^{(1)}$		

Field combinations 4 of 8

Model	α	dS_1	YS_1	dS_2	YS_2	d_F	Y_F	$S_2 H$	$S_2 F$	$S_2 \ell$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{2h\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}H_{a_1}^\dagger S_{2,j_1}^\dagger}{\epsilon}$ $\frac{-2h\overline{y_{2,i_1,r_1}}\overline{V_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}H_{a_1}^\dagger S_{2,j_1}^\dagger}{+H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}\lambda_{H2}^{(1)}$	$-C_{S_2^1}S_2^1h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}^\dagger$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_1}y_{2,y_{2,i_2,r_2}}S_{2,j_1}^\dagger S_{2,j_2}^\dagger$ $-h\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}^\dagger S_{2,j_2}^\dagger\mathcal{I}_{S2,j_1,j_3,j_2,j_4}}\lambda_{S2}^{(1)}$	$+\frac{3h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}g_{S2}^2}{8\epsilon M_F}$ $+\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}g_{S2}^2}{8M_F}$ $+\frac{3h\overline{L_{a_1,i_1}}C P_R L_{a_2,i_2}^T y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}S_{2,j_1}^\dagger S_{2,j_2}^\dagger g_{S2}^2}{8\epsilon M_F}$ $+\frac{h\overline{L_{a_1,i_1}}C P_R L_{a_2,i_2}^T y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}S_{2,j_1}^\dagger S_{2,j_2}^\dagger g_{S2}^2}{8M_F}$
cont.								$\frac{h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}{\epsilon}\lambda_{H2}^{(1)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}\lambda_{H2}^{(1)}$ $+H_{a_2}S_{2,j_2}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}\lambda_{H2}^{(2)}$		$+\frac{3h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_{Y^2}^2S_{2,j_1}S_{2,j_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}}{8\epsilon M_F}$ $+\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}g_{Y^2}^2S_{2,j_1}S_{2,j_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}}{8M_F}$ $+\frac{\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}}{2M_F}$
cont.								$\frac{h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}{\epsilon}\lambda_{H2}^{(2)}$ $-\frac{1}{2}h\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}^\dagger S_{2,j_1}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}\lambda_{H2}^{(2)}$		$-\frac{h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}y_{2,i_3,r_3}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1,i_1,r_1}}y_{2,i_2,r_2}y_{2,i_3,r_3}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}}{8M_F}$ $-\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}y_{2,i_3,r_3}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}}{4\epsilon M_F}$ $-\frac{3h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}y_{2,i_3,r_3}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}}{8M_F}$ $-\frac{h\overline{y_{2,i_1,r_1}}y_{2,i_2,r_2}y_{2,i_3,r_3}L_{a_2,i_2}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}}{4M_F}$
cont.										$+\frac{3h\overline{L_{a_1,i_1}}C P_R L_{a_2,i_2}^T g_{Y^2}^2y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{8\epsilon M_F}$ $+\frac{h\overline{L_{a_1,i_1}}C P_R L_{a_2,i_2}^T g_{Y^2}^2y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{8M_F}$ $+\frac{\overline{L_{a_1,i_1}}C P_R L_{a_2,i_2}^T y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{2M_F}$
cont.										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R L_{a_2,i_2}^T y_{2,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{2\epsilon M_F}$ $-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R L_{a_2,i_2}^T y_{2,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{4M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R L_{a_2,i_2}^T y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R L_{a_2,i_2}^T y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}\mathcal{I}_{y_{2,a_1,j_1}}\mathcal{I}_{y_{2,a_2,j_2}}S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{8M_F}$
cont.										$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R L_{a_2,i_2}^T y_{$

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell e$	$S_1 S_2 H$	$S_1 S_2 F$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{\hbar y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8} \frac{\hbar y_{1,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8} \frac{\hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\hbar \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8} \frac{\hbar y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+\frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8} \frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	$-\frac{2\hbar y_{2,i_1,r_1} \overline{Y_{e,i_2,i_3} H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \mathcal{I}_{y_{y_1,a_1,j_1} \mathcal{I}_{y_{2,a_2,j_2}} H_{a_1}^\dagger}}{\epsilon}$ $-2\hbar y_{2,i_1,r_1} \overline{Y_{e,i_2,i_3} H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} \mathcal{I}_{y_{y_1,a_1,j_1} \mathcal{I}_{y_{2,a_2,j_2}} H_{a_1}^\dagger}}{\epsilon}$ $+\lambda \overline{\mathcal{I}_{T3} S_{2,j_2} \mathcal{I}_{T3}^* a_{1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}$ $-\frac{\hbar y_{1,i_1,r_1} \overline{\lambda \mathcal{I}_{T3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3}^* a_{1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}{2\epsilon}$ $-\frac{1}{4} \frac{\hbar y_{1,i_1,r_1} \overline{\lambda \mathcal{I}_{T3} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3}^* a_{1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}{4\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} \overline{\lambda \mathcal{I}_{T3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3}^* a_{1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}{2\epsilon}$ $-\frac{1}{4} \frac{\hbar y_{2,i_1,r_1} \overline{\lambda \mathcal{I}_{T3} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3}^* a_{1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger}}{4\epsilon}$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda \mathcal{I}_{T3} S_{2,j_1}^\dagger$ $-\frac{\hbar y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda \mathcal{I}_{T3} S_{2,j_1}^\dagger}{2\epsilon}$ $-\frac{1}{4} \frac{\hbar y_{1,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda \mathcal{I}_{T3} S_{2,j_1}^\dagger}{4\epsilon}$ $\frac{\hbar y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda \mathcal{I}_{T3} S_{2,j_1}^\dagger}{2\epsilon}$ $-\frac{1}{4} \frac{\hbar y_{2,i_1,r_1} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda \mathcal{I}_{T3} S_{2,j_1}^\dagger}{4\epsilon}$ $\frac{2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3} H_{a_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} \mathcal{I}_{y_{y_2,a_1,j_2} \mathcal{I}_{y_{1,a_2,j_1}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}{\epsilon}$ $-2\hbar y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3} H_{a_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} \mathcal{I}_{y_{y_2,a_1,j_2} \mathcal{I}_{y_{1,a_2,j_1}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger}}{\epsilon}$ $+\frac{\hbar y_{2,i_1,r_1} \overline{H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_{y_1,a_1,j_1} \mathcal{I}_{y_{2,a_2,j_2}} H_{a_1}^\dagger \lambda_{H2}^{(1)}}}}{\epsilon}$ $+\frac{1}{2} \frac{\hbar y_{2,i_1,r_1} H_{a_2} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_{y_1,a_1,j_1} \mathcal{I}_{y_{2,a_2,j_2}} H_{a_1}^\dagger \lambda_{H2}^{(1)}}}{\epsilon}$ $+\frac{\hbar y_{1,i_1,r_1} H_{a_2} y_{2,i_1,r_1} \mathcal{I}_{y_{2,a_1,j_1} \mathcal{I}_{y_{1,a_2,j_2}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(1)}}}{\epsilon}$ $+\frac{1}{2} \frac{\hbar y_{1,i_1,r_1} H_{a_2} y_{2,i_1,r_1} \mathcal{I}_{y_{2,a_1,j_1} \mathcal{I}_{y_{1,a_2,j_2}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(1)}}}{\epsilon}$ $+\frac{\hbar y_{2,i_1,r_1} H_{a_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_1,j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$ $+\frac{\hbar y_{2,i_1,r_1} H_{a_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_1,j_2} H_{a_1}^\dagger \lambda_{H2}^{(2)}}{2\sqrt{3}}$ $-\frac{\hbar y_{2,i_1,r_1} H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{y_{y_1,a_1,j_1} \mathcal{I}_{y_{2,a_2,j_2}} H_{a_1}^\dagger \lambda_{H2}^{(2)}}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{2,i_1,r_1} H_{a_2} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{y_{y_1,a_1,j_1} \mathcal{I}_{y_{2,a_2,j_2}} H_{a_1}^\dagger \lambda_{H2}^{(2)}}}{2\sqrt{3}}$ $+\frac{\hbar y_{1,i_1,r_1} \text{cg}1_{j_1,j_2} H_{a_1} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}}{\sqrt{3}\epsilon}$ $+\frac{\hbar y_{1,i_1,r_1} \text{cg}1_{j_1,j_2} H_{a_1} y_{2,i_1,r_1} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}}{2\sqrt{3}}$ $-\frac{\hbar y_{1,i_1,r_1} H_{a_2} y_{2,i_1,r_1} \mathcal{I}_{y_{2,a_1,j_2} \mathcal{I}_{y_{1,a_2,j_1}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}}}{\sqrt{3}\epsilon}$ $-\frac{\hbar y_{1,i_1,r_1} H_{a_2} y_{2,i_1,r_1} \mathcal{I}_{y_{2,a_1,j_2} \mathcal{I}_{y_{1,a_2,j_1}} H_{a_1}^\dagger S_{1,j_1}^\dagger S_{2,j_2}^\dagger \lambda_{H2}^{(2)}}}{2\sqrt{3}}$	$+C_{S_1^2 S_2} \hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right)}{\overline{y_{1,i_1,r_1} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} \text{Bar}(\text{cg}1)_{j_1,j_2}}}$ $-\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \frac{\hbar \log \left(\frac{\bar{\mu}^2}{M_F^2} \right)}{\overline{y_{2,i_1,r_1} y_{2,i_2,r_2} S_{1,j_1} S_{1,j_2} S_{2,j_2} S_{2,j_3} y_{1,i_1,r_1} y_{1,i_2,r_2} y_{1,i_3,r_3} \text{Bar}(\text{cg}1)_{j_1,j_2} \text{Bar}(\text{cg}1)_{j_3$

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[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 S_2 \ell$	$S_1 F H$	$S_1 F \ell$
cont.								$+ \frac{\hbar \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$		
								$+ \frac{\hbar \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$		
								$+ \frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$		
cont.								$+ \frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}^\dagger \lambda_{12}^{(1)}}{M_F}$		
								$- \frac{\hbar \overline{y_{1,i_2,r_1}} y_{2,i_1,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_1,a_1,j_1}^* \mathcal{I}_{y_2,a_2,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} \epsilon M_F}$		
								$- \frac{\hbar \overline{y_{1,i_2,r_1}} y_{2,i_1,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_1,a_1,j_1}^* \mathcal{I}_{y_2,a_2,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$		
cont.								$- \frac{\hbar \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_1,a_1,j_1}^* \mathcal{I}_{y_2,a_2,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} \epsilon M_F}$		
								$- \frac{\hbar \overline{y_{1,i_1,r_1}} y_{2,i_2,r_1} L_{a_2,i_2}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{y_1,a_1,j_1}^* \mathcal{I}_{y_2,a_2,j_2} S_{1,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$		
								$- \frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} \epsilon M_F}$		
cont.								$- \frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$		
								$- \frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} \epsilon M_F}$		
								$- \frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$		

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	F	H	S_2	F	ℓ	F	H	ℓ
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} y_{e,i_1,i_3} I_{y_2,a_1,j_1} \overline{I_{y_2,a_2,j_2}} H_{a_1}^\dagger S_{2,j_1}^\dagger$			$\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} \overline{I_{y_2,a_1,j_1}} \overline{I_{y_2,a_2,j_2}} y_{2,i_2}^2}{8M_F}$			$\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_3} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$		
								$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H2,a_1,a_2,j_1,j_2}^\dagger \lambda_{H2}^{(1)}$			$\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_{2,i_1,r_1} y_{2,i_2,r_1} I_{y_2,a_1,j_1} \overline{I_{y_2,a_2,j_2}} S_{2,j_1}^\dagger S_{2,j_2}^\dagger y_{2,i_2}^2}{8M_F}$			$\frac{2h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$		
								$-h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}^\dagger S_{2,j_1}^\dagger I_{H2,a_1,a_2,j_1,j_2}^\dagger \lambda_{H2}^{(2)}$			$\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} y_1^2 S_{2,j_1} S_{2,j_2} \overline{I_{y_2,a_1,j_1}} \overline{I_{y_2,a_2,j_2}}}{8M_F}$					
cont.											$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{1,i_3,r_1} \overline{I_{y_2,a_1,j_1}} \overline{I_{y_2,a_2,j_2}}}{4M_F}$					
											$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{y_2,a_1,j_1}} \overline{I_{y_2,a_2,j_2}}}{4M_F}$					
											$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} S_{2,j_2} y_{2,i_3,r_2} \overline{I_{y_2,a_1,j_1}} \overline{I_{y_2,a_2,j_2}}}{2M_F}$					
cont.											$\frac{3h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{I_{a_1,i_1}} C P_R \overline{I_{a_2,i_2}}^T y_2^2 y_{2,i_1,r_1} y_{2,i_2,r_1} \overline{I_{y_2,a_1,j_1}} \overline{I_{y_2,a_2,j_2}} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{8M_F}$					
											$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{I_{a_1,i_1}} C P_R \overline{I_{a_2,i_2}}^T y_{2,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \overline{I_{y_2,a_1,j_1}} \overline{I_{y_2,a_2,j_2}} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{2M_F}$					
											$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} \overline{I_{y_2,a_1,j_1}} \overline{I_{y_2,a_2,j_2}} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{4M_F}$					
cont.											$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} \overline{I_{y_2,a_1,j_1}} \overline{I_{y_2,a_2,j_2}} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{4M_F}$					
											$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} \overline{I_{y_2,a_1,j_1}} \overline{I_{y_2,a_2,j_2}} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{4M_F}$					
											$\frac{h \log \left(\frac{\mu^2}{M_F^2} \right) \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_{2,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2} \overline{I_{y_2,a_1,j_1}} \overline{I_{y_2,a_2,j_2}} S_{2,j_1}^\dagger S_{2,j_2}^\dagger}{2M_F}$					

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[illegible]